

Patient Information

|                 |            |                |                  |
|-----------------|------------|----------------|------------------|
| Patient ID      | RTA-2418   | Name           | Aniket Wahul     |
| Age / Sex       | 23 yrs / M | Diabetes       | 4 years duration |
| Previous DR     | Mild NPDR  | HbA1c          | —                |
| Prev. Screening | —          | Screening Date | 15 Sep 2026      |

Image Quality Assessment

| Eye   | Status | Focus | Brightness | Contrast | FOV | Enhanced |
|-------|--------|-------|------------|----------|-----|----------|
| Left  | GOOD   | 720   | 69         | 100      | 92  | No       |
| Right | GOOD   | 730   | 73         | 100      | 92  | No       |

AI Screening Results

| LEFT EYE                                    | RIGHT EYE                                 |
|---------------------------------------------|-------------------------------------------|
| <div>Grade 2</div> <div>Moderate NPDR</div> | <div>Grade 3</div> <div>Severe NPDR</div> |
| REFERABLE                                   | REFERABLE                                 |
| Image Quality: Good                         | Image Quality: Good                       |

Left Eye — Grade Probability Distribution

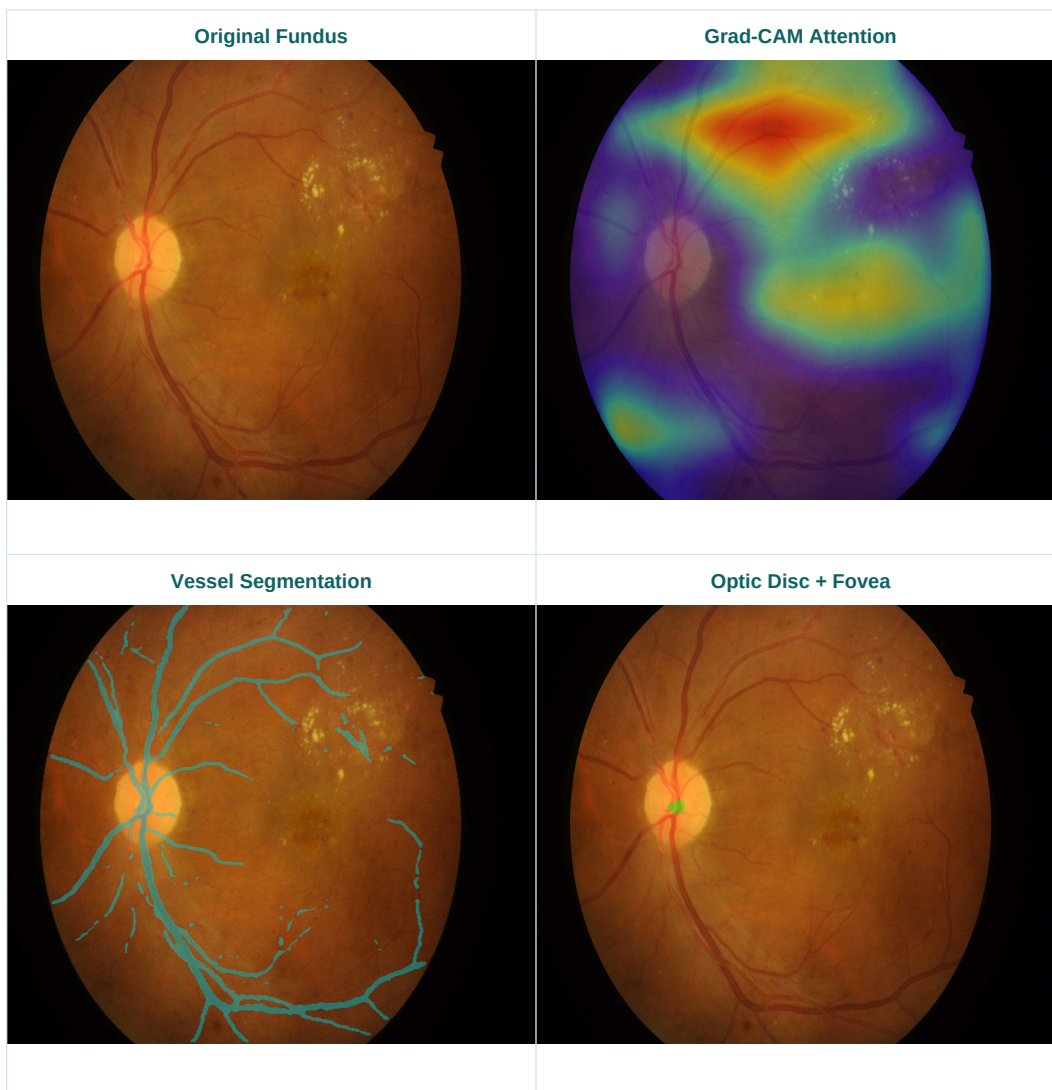
|                         |  |       |
|-------------------------|--|-------|
| Grade 0 — No DR         |  | 0.5%  |
| Grade 1 — Mild          |  | 4.8%  |
| Grade 2 — Moderate      |  | 92.1% |
| Grade 3 — Severe        |  | 1.4%  |
| Grade 4 — Proliferative |  | 1.2%  |

Right Eye — Grade Probability Distribution

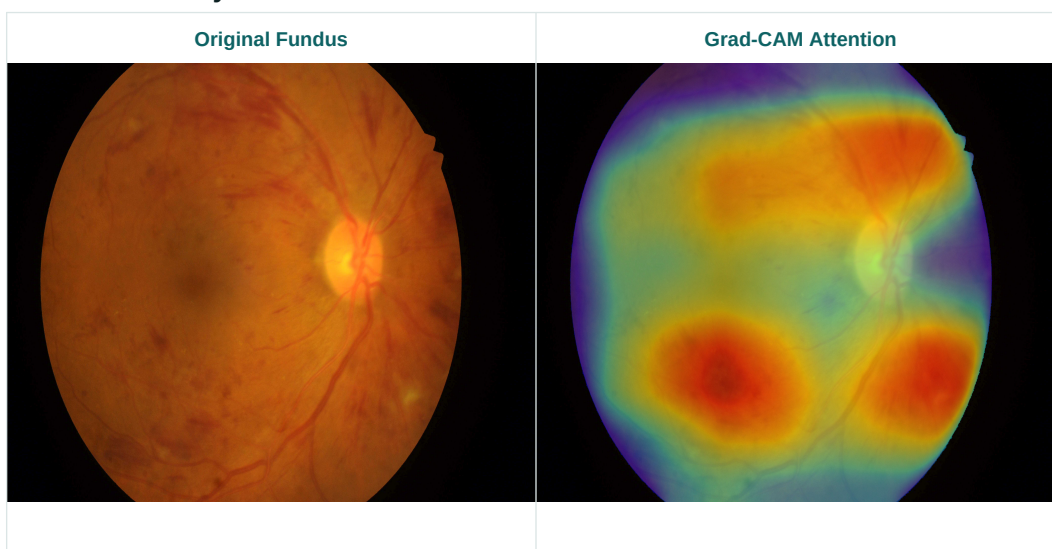
|                         |  |       |
|-------------------------|--|-------|
| Grade 0 — No DR         |  | 0.1%  |
| Grade 1 — Mild          |  | 0.1%  |
| Grade 2 — Moderate      |  | 0.8%  |
| Grade 3 — Severe        |  | 86.2% |
| Grade 4 — Proliferative |  | 12.8% |

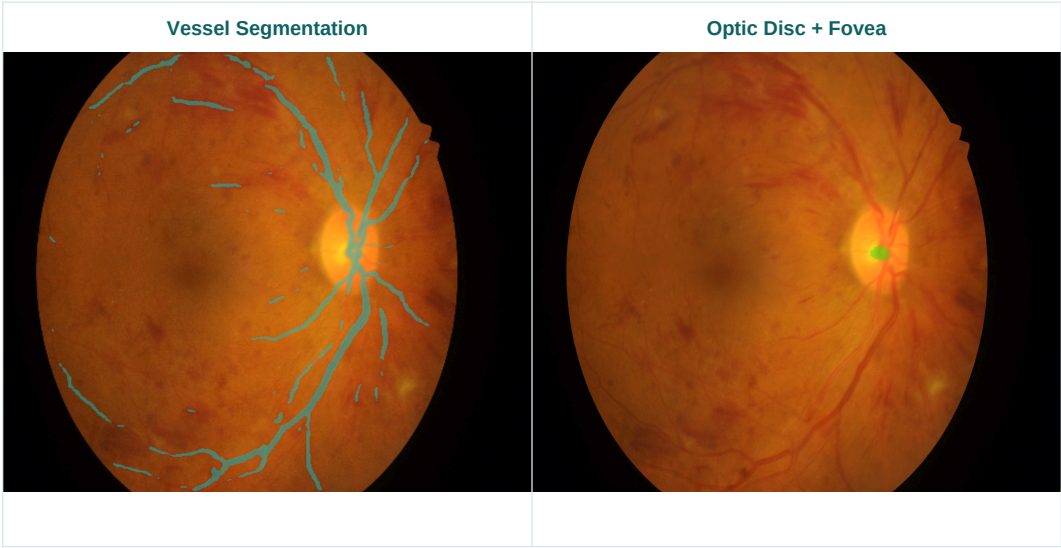
## Visual Evidence

### Left Eye — Retinal Analysis



### Right Eye — Retinal Analysis





### Lesion Analysis & Microvascular Pathology

#### Left Eye — Microvascular Lesion Breakdown

Left Eye · Lesion Overlay

| Lesion Type              | Map Color | Status       | Count |
|--------------------------|-----------|--------------|-------|
| Microaneurysms           | Blue      | Detected     | 44    |
| Intraretinal Hemorrhages | Orange    | Detected     | 13    |
| Hard Exudates            | Yellow    | Detected     | 56    |
| Neovascularization       | Clear     | Not detected | 0     |

● High microvascular lesion load

Total 113 focal lesions quantified via 5-fold ensemble.

• **Macular Advisory:** Hard exudates present — evaluate foveal proximity for DME risk.

Key:

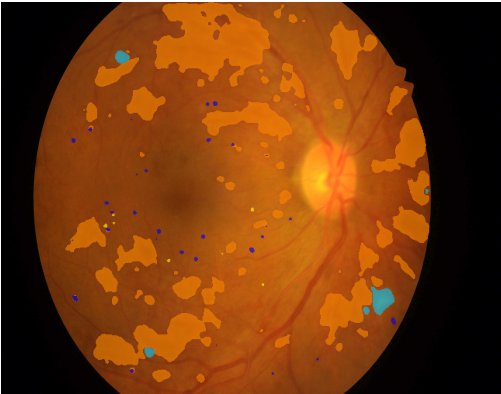
● Microaneurysms

● Hemorrhages

● Exudates

#### Right Eye — Microvascular Lesion Breakdown

Right Eye · Lesion Overlay



| Lesion Type              | Map Color | Status       | Count |
|--------------------------|-----------|--------------|-------|
| Microaneurysms           | Blue      | Detected     | 31    |
| Intraretinal Hemorrhages | Orange    | Detected     | 73    |
| Hard Exudates            | Yellow    | Detected     | 12    |
| Neovascularization       | Clear     | Not detected | 0     |

● High microvascular lesion load

Total **116** focal lesions quantified via 5-fold ensemble.

• **Macular Advisory:** Hard exudates present — evaluate foveal proximity for DME risk.

Key:

● Microaneurysms

■ Hemorrhages

■ Exudates

Referral Recommendation

PRIORITY REFERRAL

Patient requires immediate referral to an ophthalmologist for detailed clinical evaluation and management.

Ophthalmologist Review

|                   |                       |
|-------------------|-----------------------|
| Reviewer          | Dr. Smith             |
| Decision          | ↘ Assessment Modified |
| Final Grade (L/R) | Grade 2 / Grade 3     |
| Final Referral    | Referable             |
| Notes             | —                     |
| Reviewed at       | 15 Sep 2026, 15:08    |

AI pipeline completed in 136.3 seconds.

DISCLAIMER: This report is generated by the RetinaAI AI-assisted screening system (SIH 26038) and is intended for use as a clinical decision-support tool only. AI outputs are not a substitute for professional medical examination. Final clinical decisions must be made by a qualified healthcare professional. Grad-CAM attention maps indicate model focus regions and are not equivalent to lesion segmentation masks.